Spin-Orbit Torque and Nernst Effects in Bi_xSb_{1-x}/Ferromagnet Heterostructures

Niklas Roschewsky, Praveen Gowtham, Frances Hellman, Sayeef Salahuddin

University of California, Berkeley

Emily Walker, Seth R. Bank

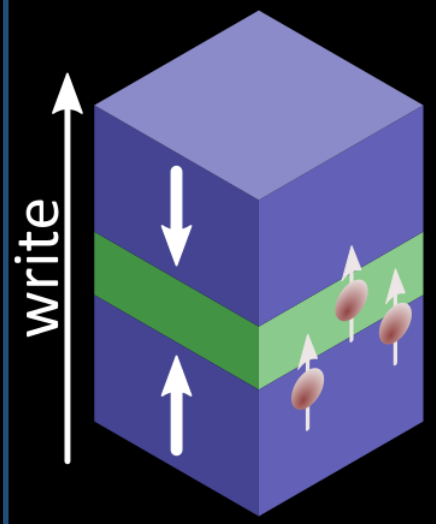
University of Texas, Austin

AB-07

Berkeley
UNIVERSITY OF CALIFORNIA

Spin-torque memory

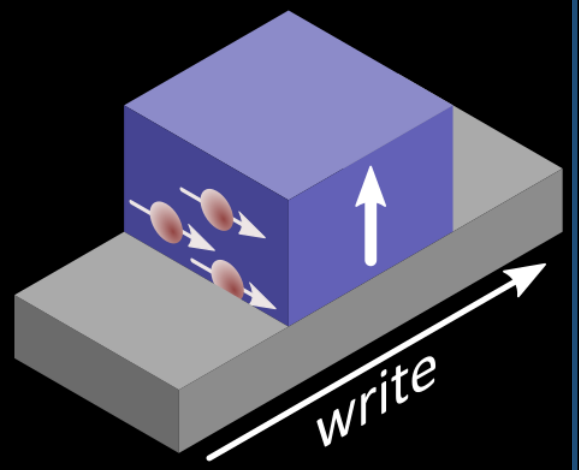
Spin Transfer Torque - MRAM



- Fast
- Reliable
- High efficiency j_c/Δ

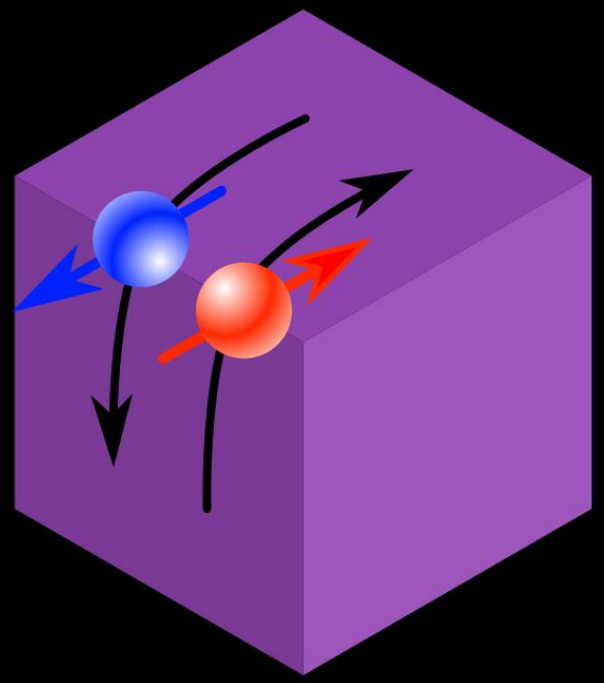
Spin Orbit Torque - MRAM

- Faster
- Better Reliability
- Low efficiency j_c/Δ



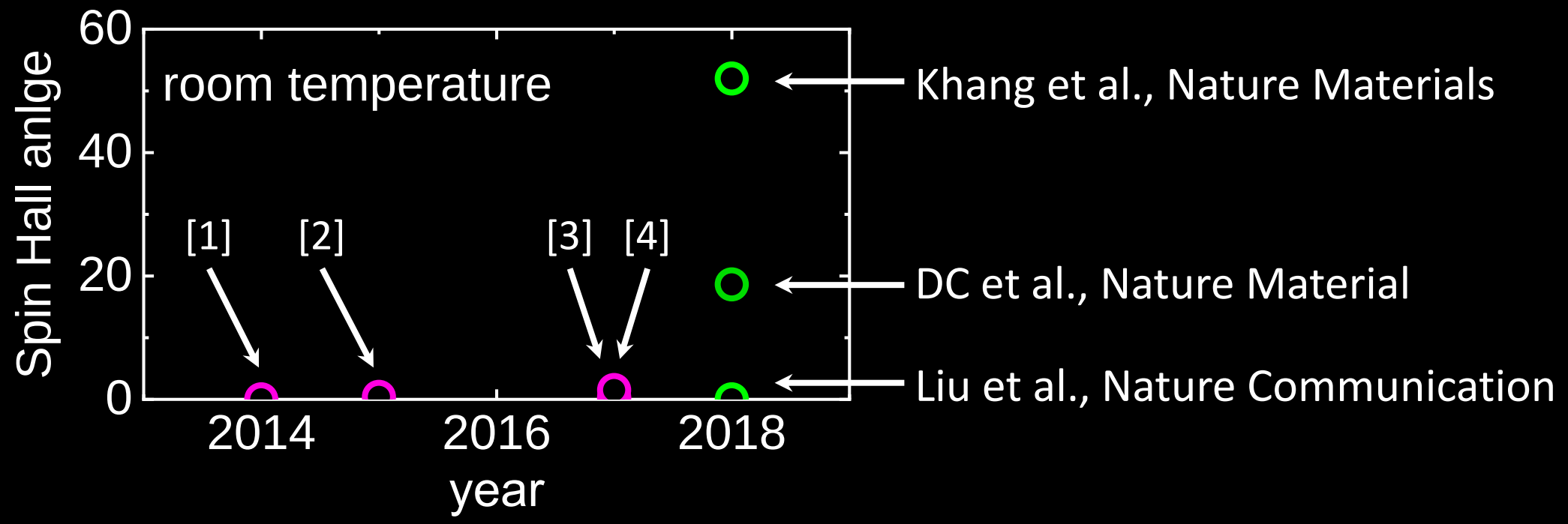
Is low SOT efficiency j_c/Δ a materials challenge?

Spin Torque in topological Insulators



Ultra large spin-Hall angle in topological insulators

Spin Hall Angle in Topological Insulators

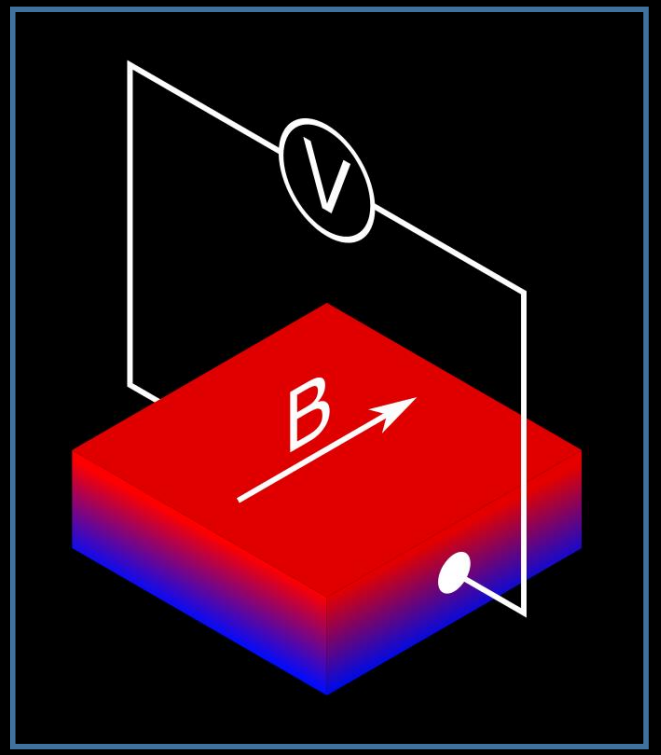


[1] Deorani et al., Physical Review B
[2] Jamali et al., Nano Letters
[3] Han et al., Physical Review Letters
[4] Wang et al., Nature Communications

Ordinary Nernst Effect in Topological Insulators

The ordinary Nernst effect is large in topological insulators

This might lead to an overestimation of the spin Hall angle



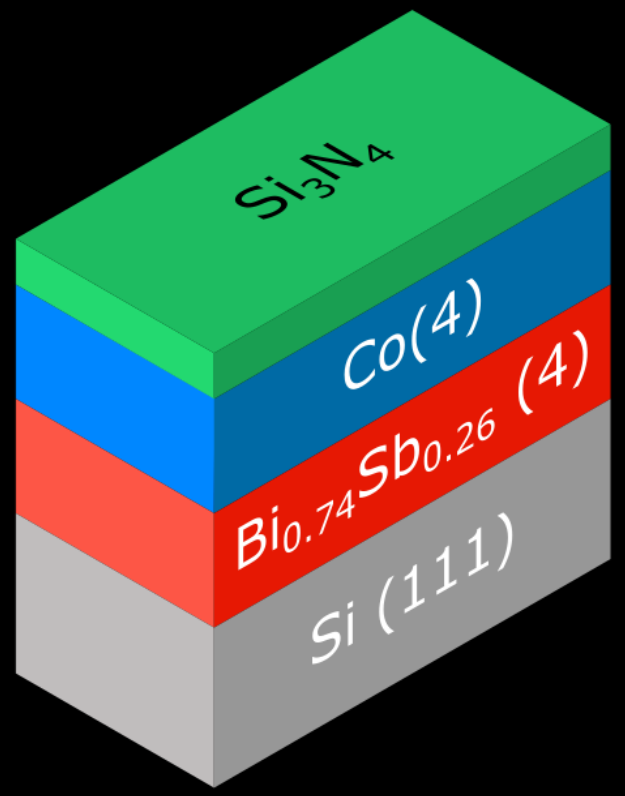
Overview

Harmonic Hall voltage persists in control samples without magnet

Harmonic Hall voltage is not suppressed at large magnetic field

A longitudinal electric field of same magnitude is generated

Sample overview

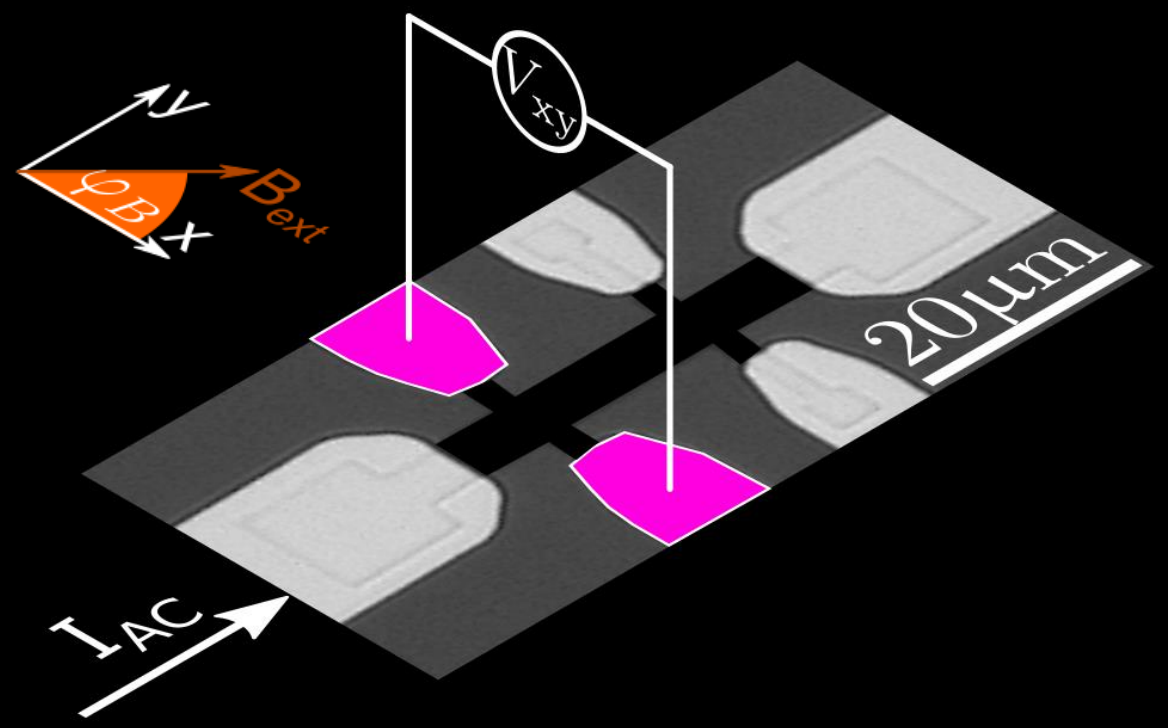


- Thermal Effects
- Spin-orbit torque

- Thermal Effects
- No Spin-orbit torque

- Spin-orbit torque
- “Small” thermal effects

Harmonic Hall Measurements of SOT



Hall Voltage V_{xy}

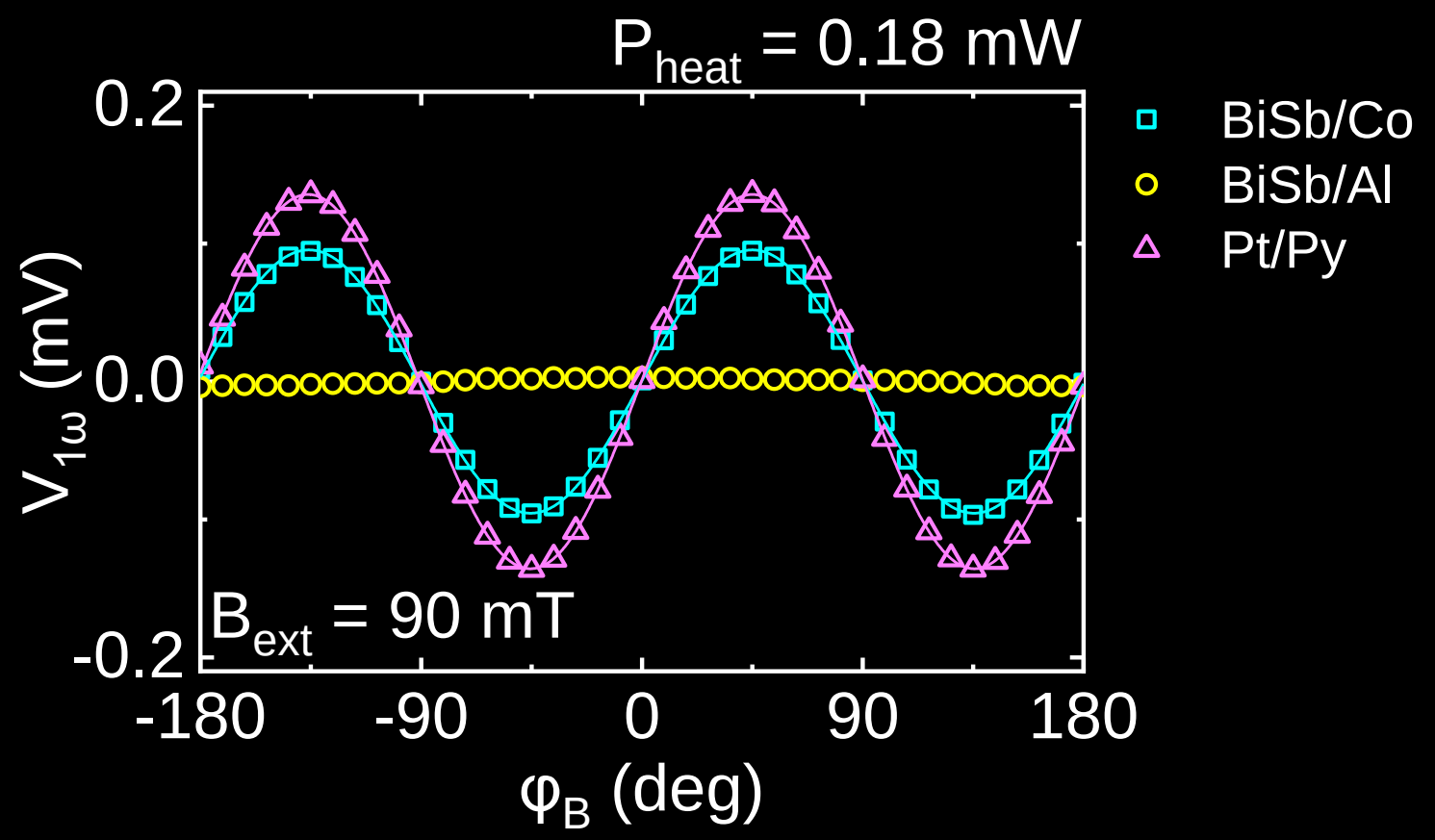
First Harmonic:

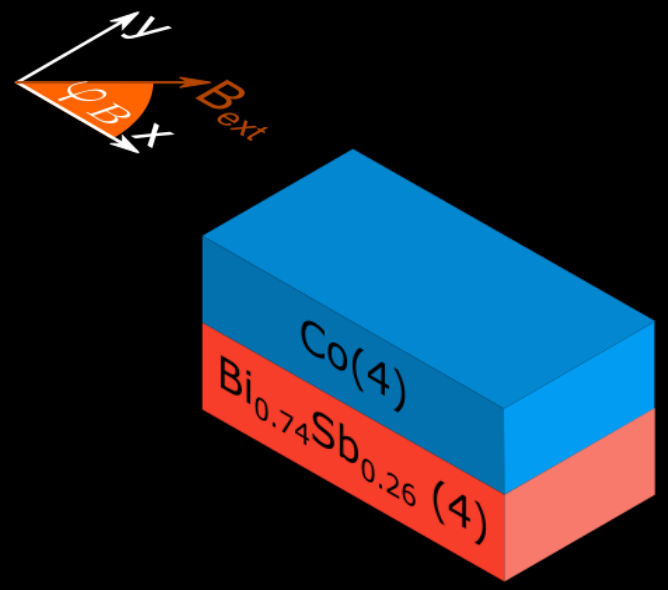
- Planar Hall Effect

Second Harmonic:

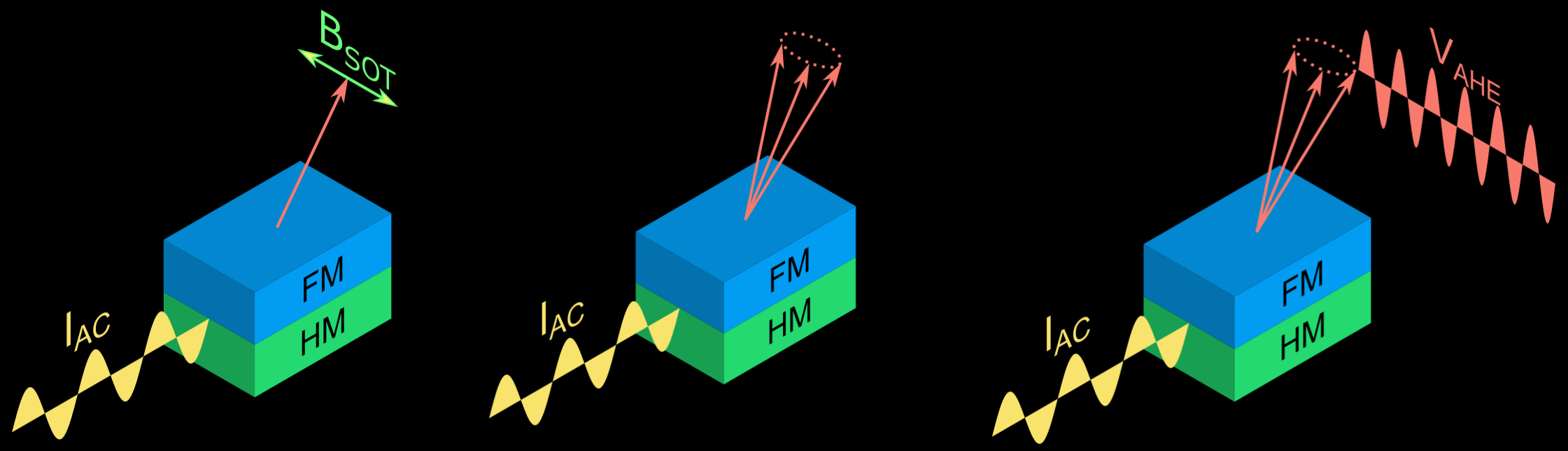
- Spin orbit torque
- Thermal Effects

Planar Hall Effect



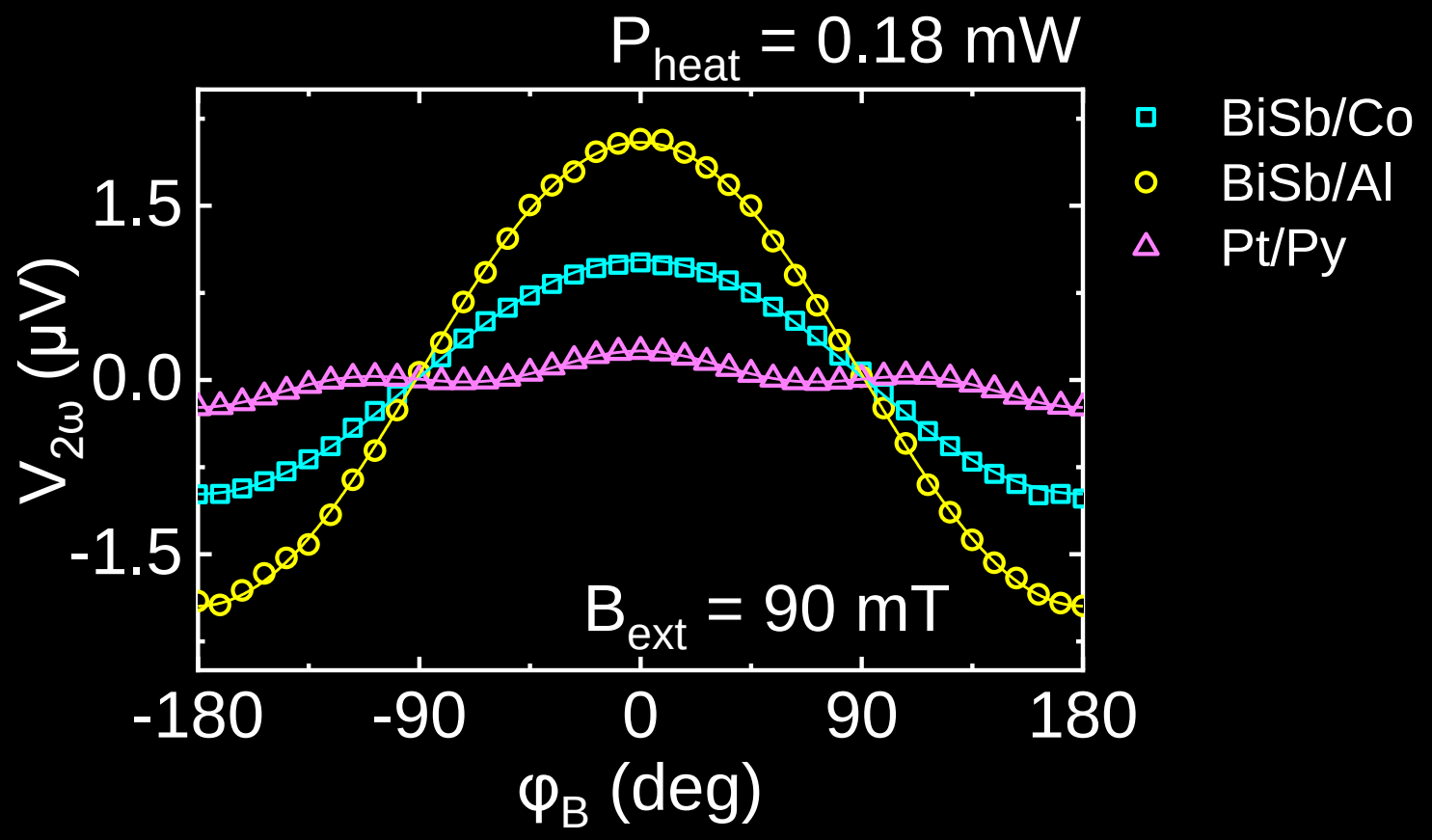


Second Harmonic Hall voltage



I_{AC} induces B_{SOT} → M oscillates with B_{SOT} → V_{xy} changes periodically

Spin Orbit Torque



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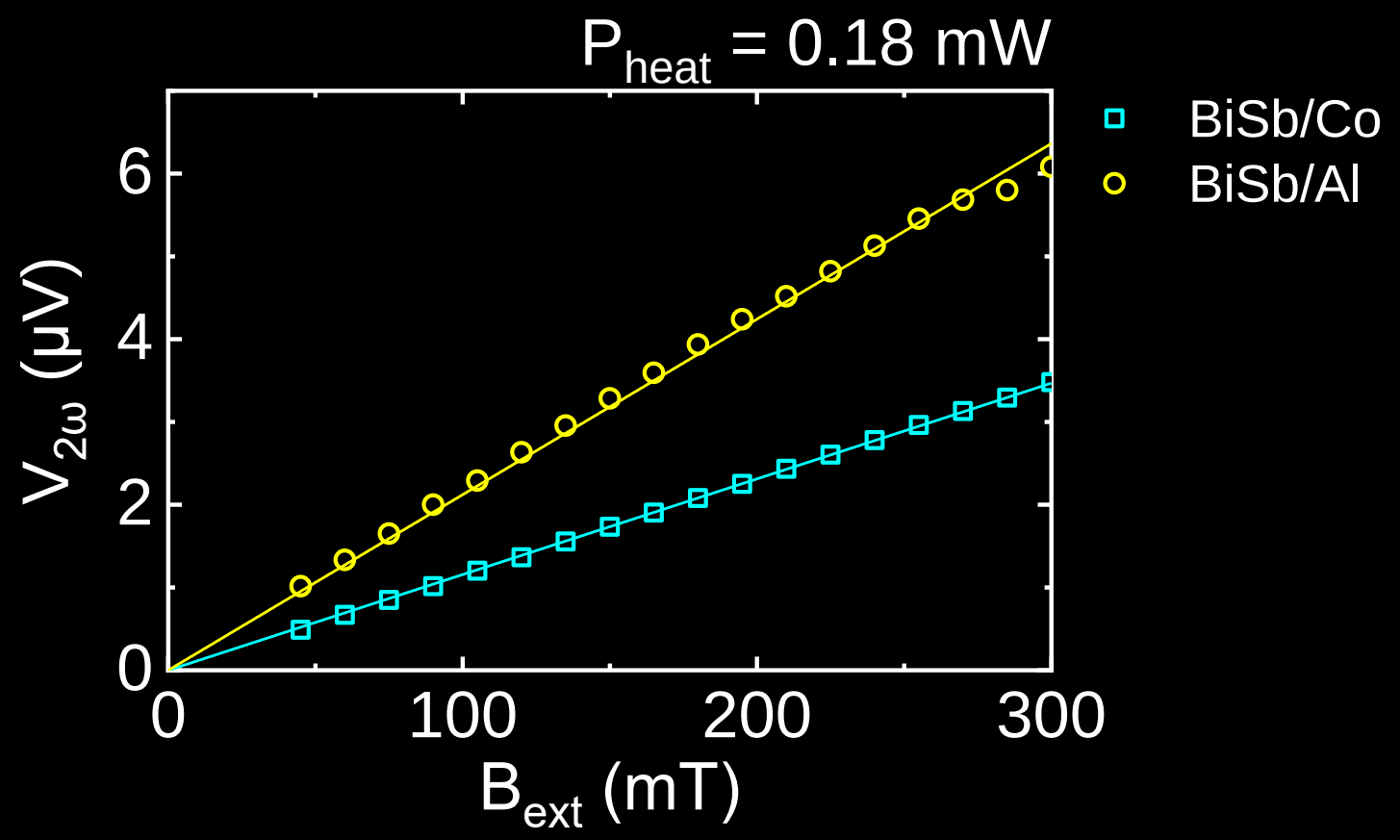
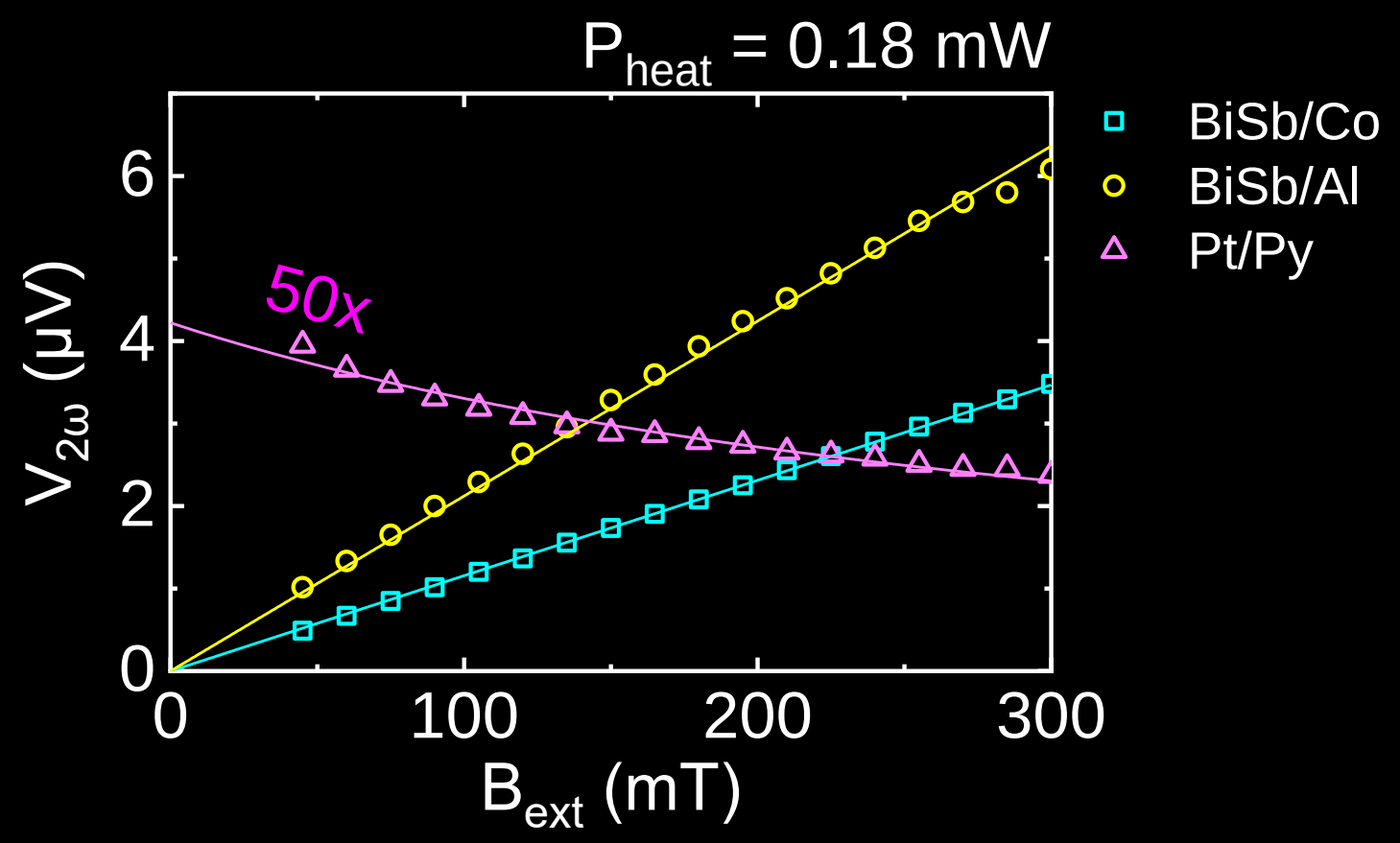
A longitudinal electric field of same magnitude is generated

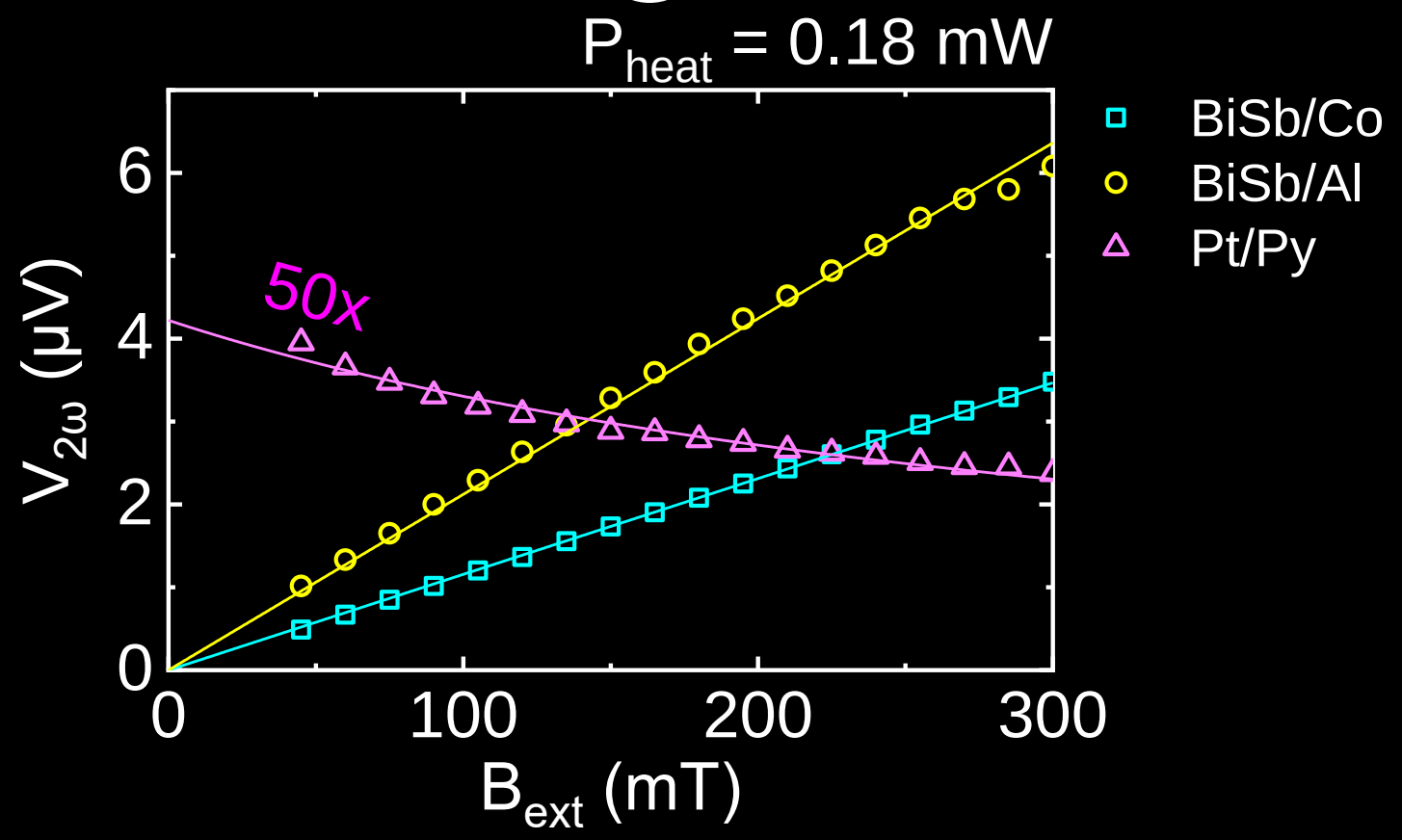
Magnetic Field Scaling



Magnetic Field Scaling



Magnetic Field Scaling



Linear scaling confirms presence of ordinary Nernst effect

Overview

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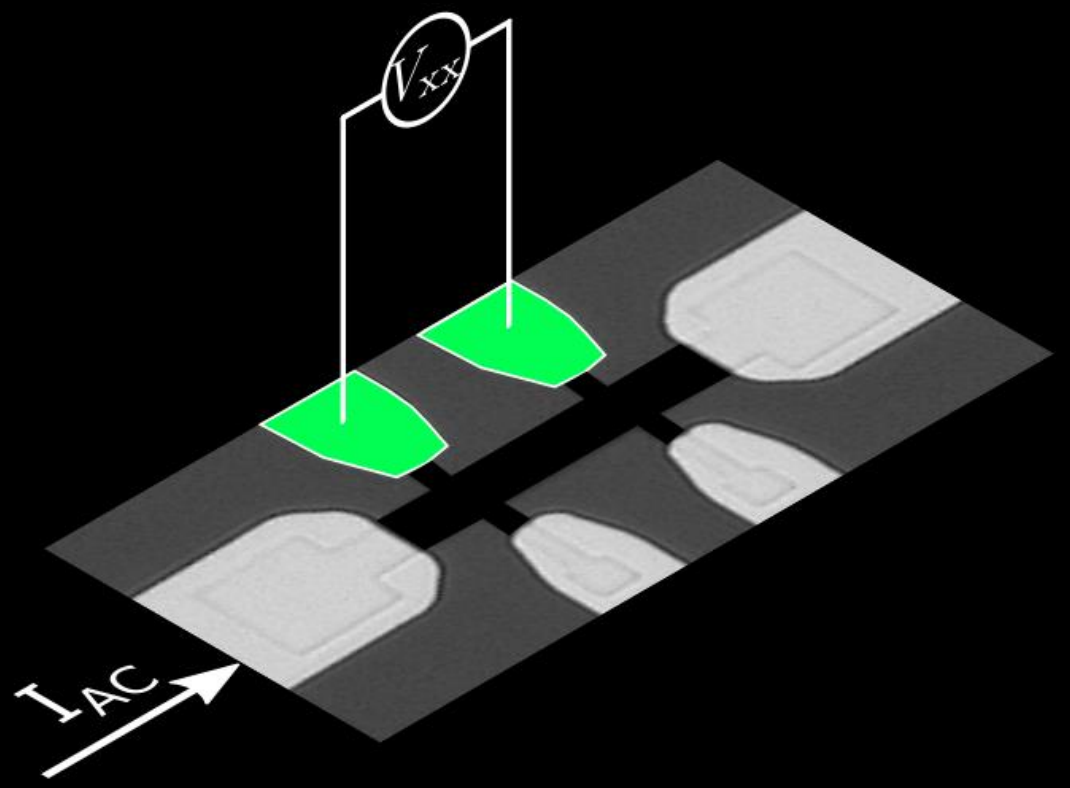
A longitudinal electric field of same magnitude is generated

Longitudinal Voltage



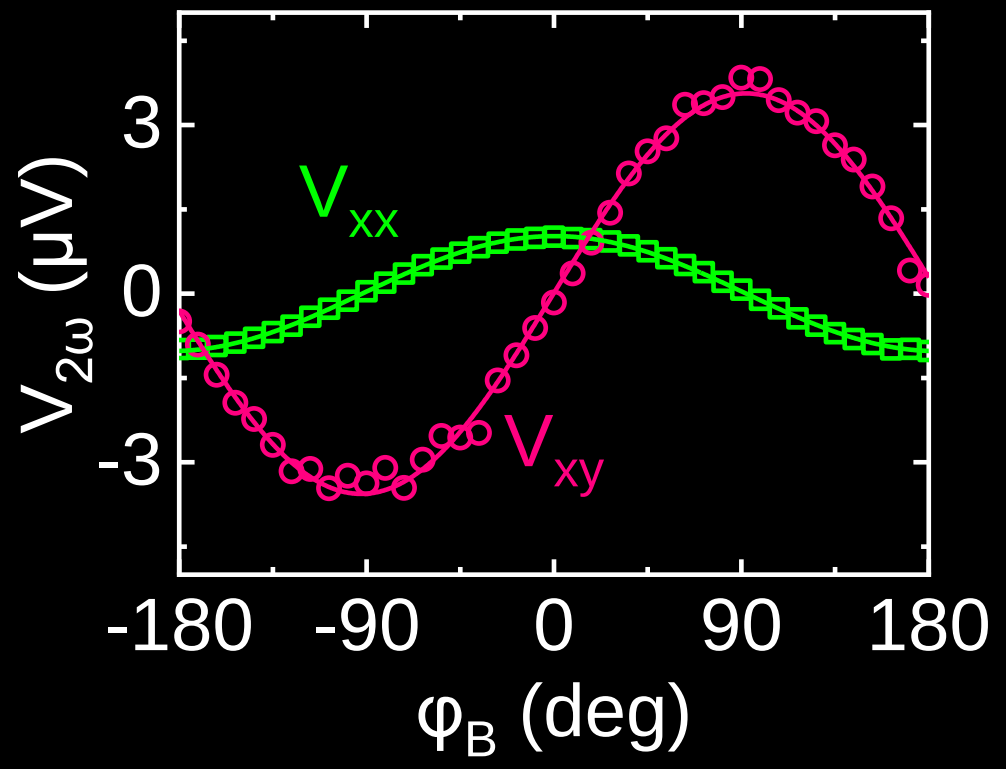
First Harmonic:

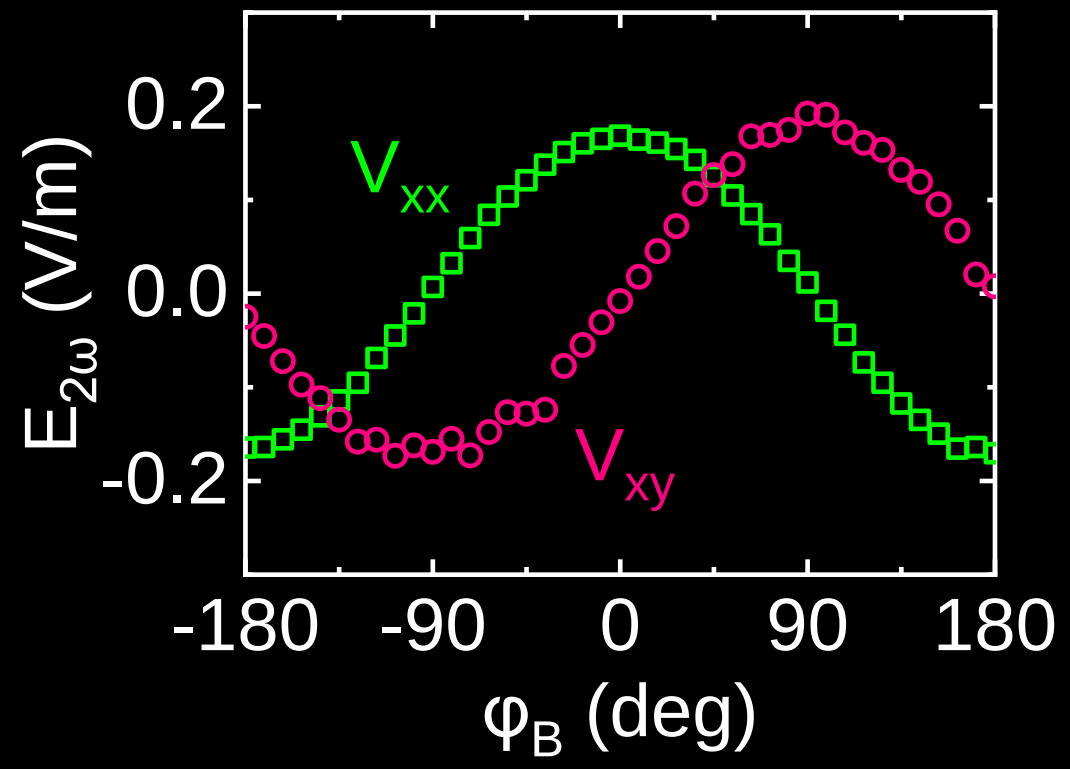
- AMR

Second Harmonic:

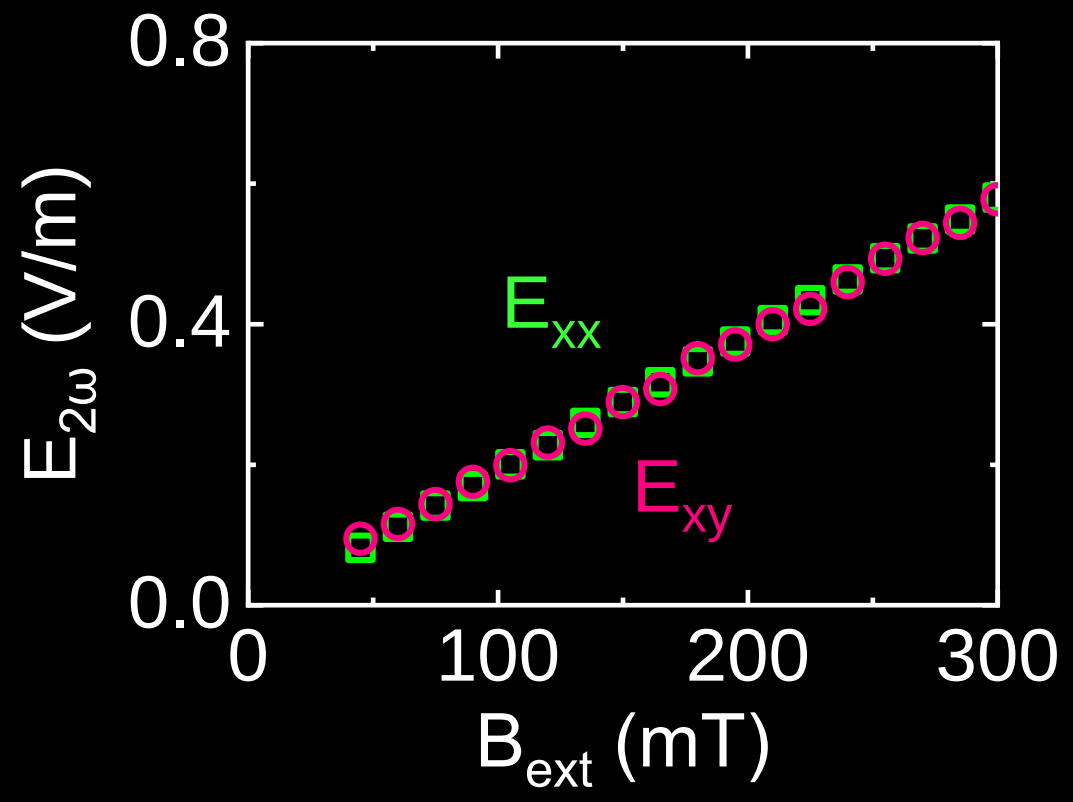
- USMR
- Thermal Effects
- No damping like SOT

Thermal voltages Bi-Sb/Co

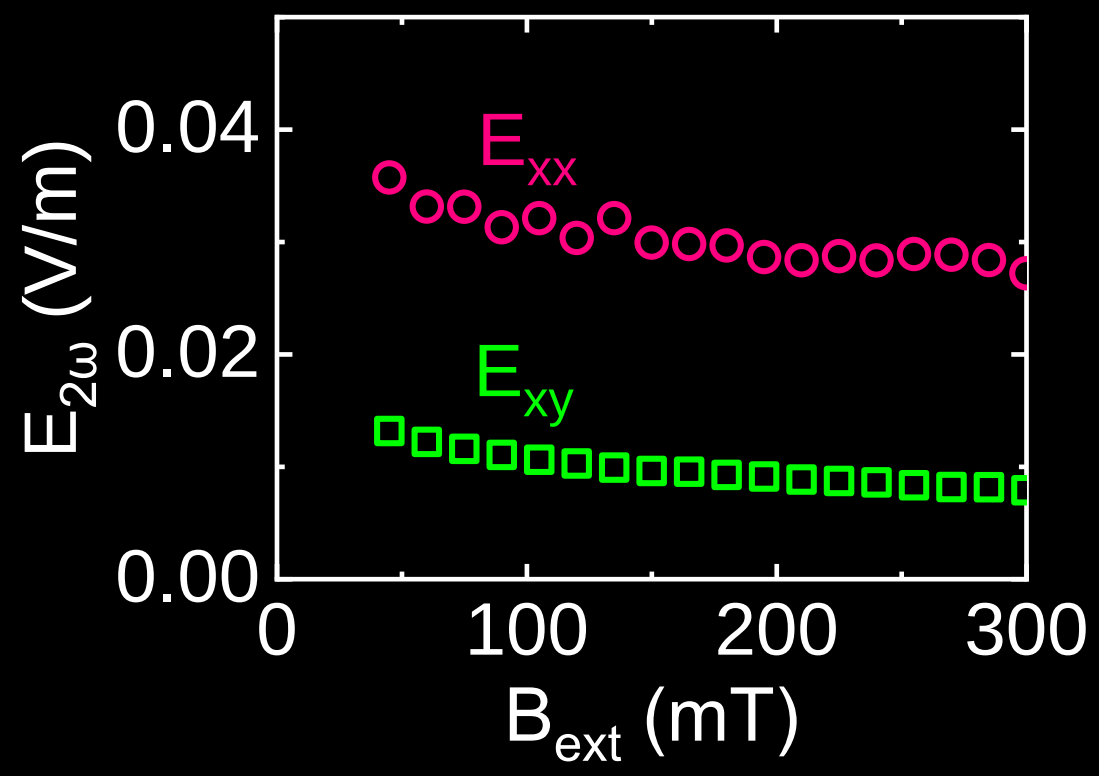




Spin Orbit Torque - Electric field



Bi-Sb/Co



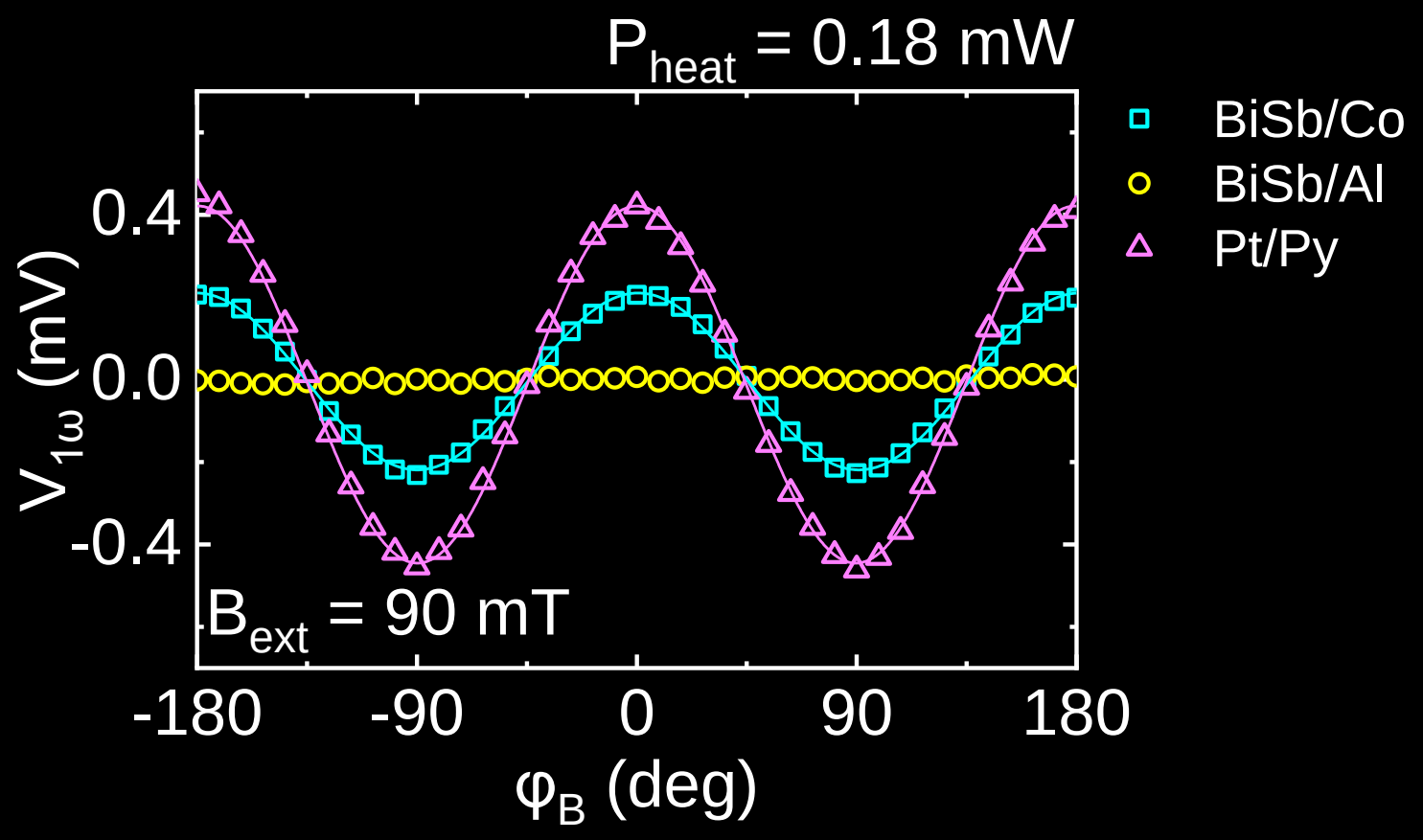
Pt/Py

Appendix

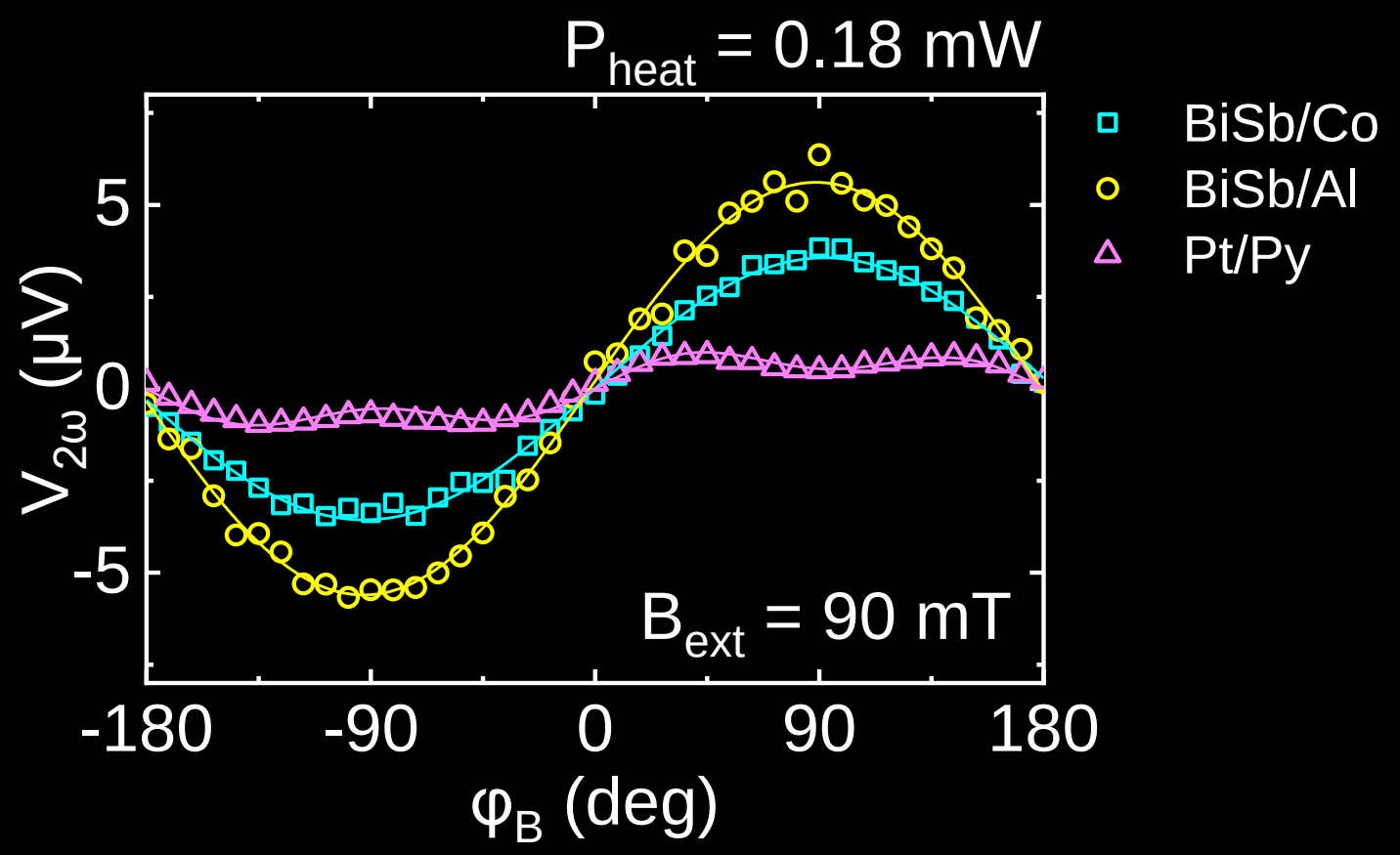
Do do

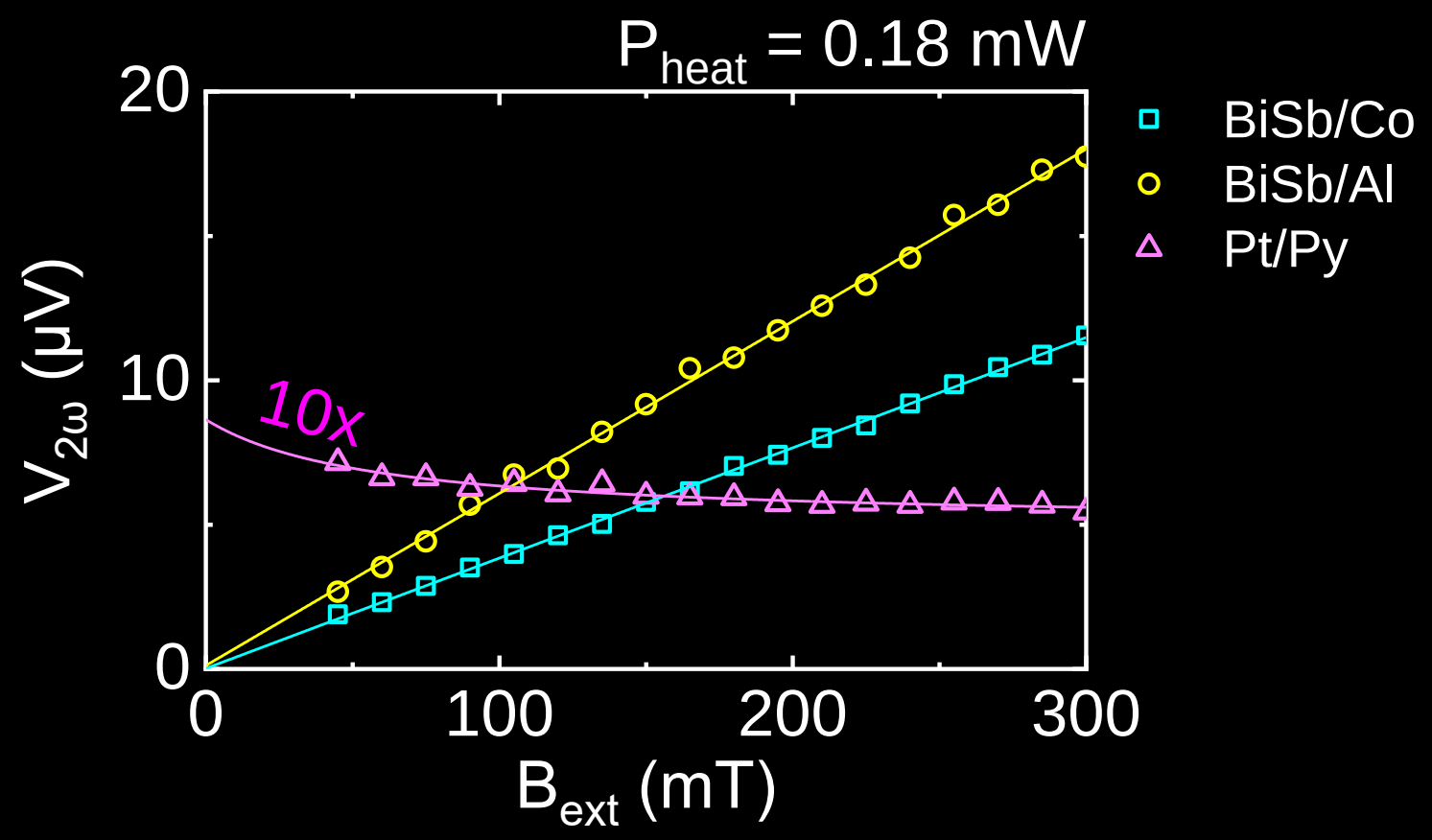
- Introduce samples/ thicknesses

Anisotropic Magnetoresistance



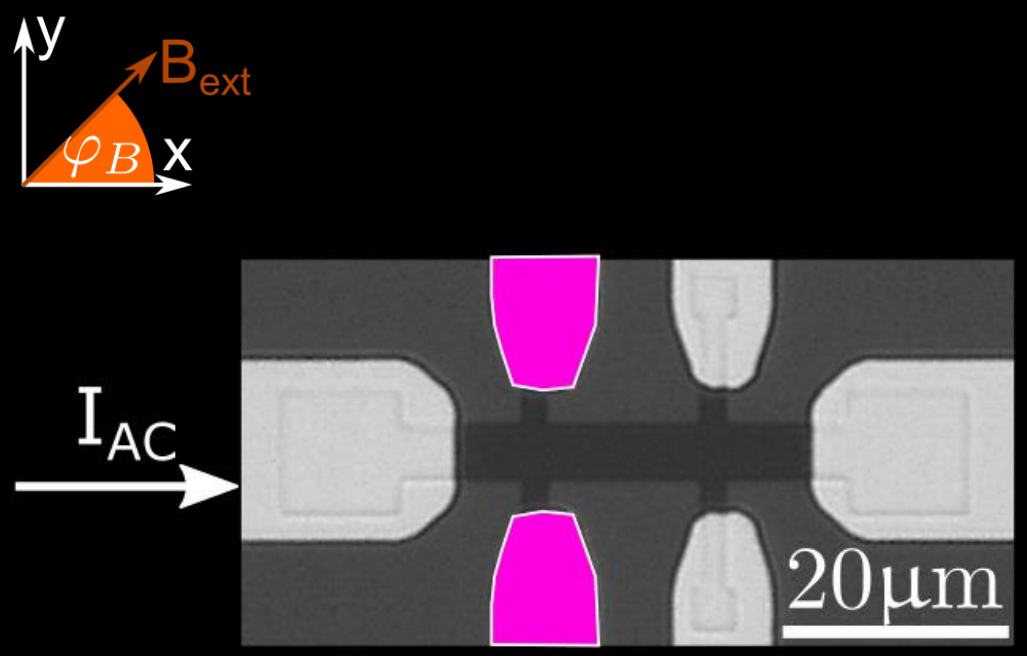
Unidirectional Spin Hall Magnetoresistance



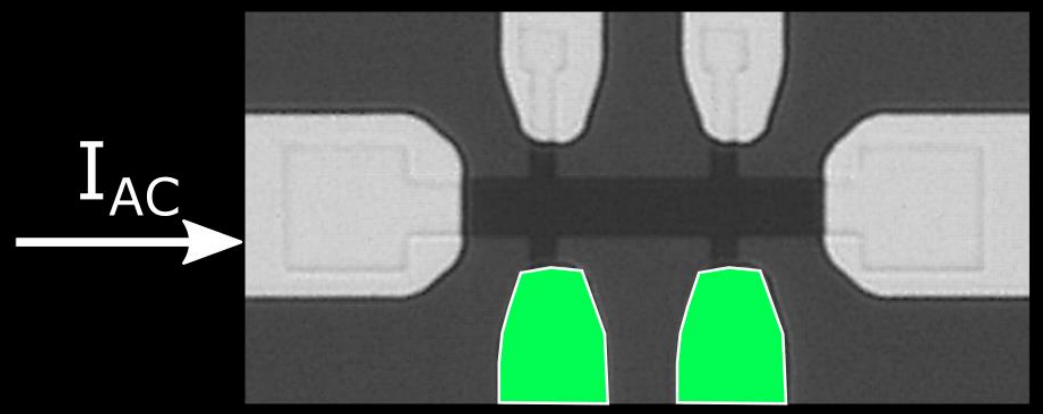

P_{heat} = 0.18 mW
20
10
0
0
100
200
300
$V_{2\omega}$ (μV)
B_{ext} (mT)
10x
BiSb/Co
BiSb/Al
Pt/Py

Old Slides

Harmonic Hall Measurements of SOT



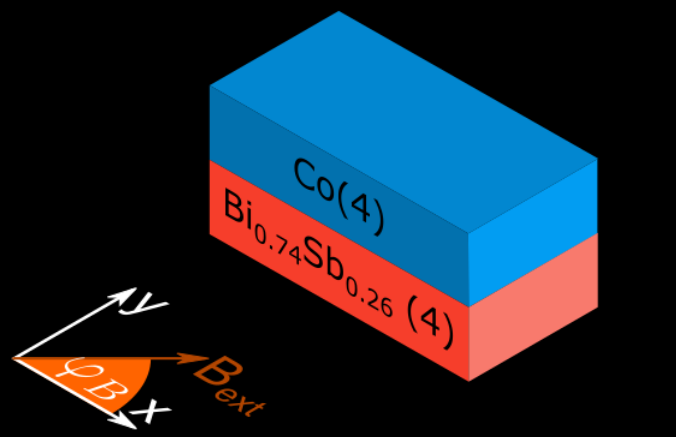
Hall Voltage V_{xy}

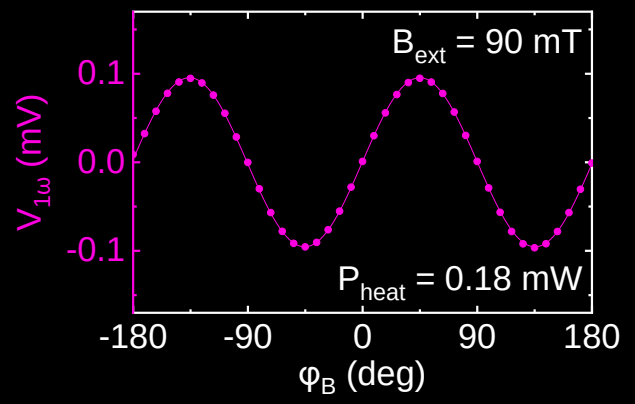


Longitudinal Voltage V_{xx}

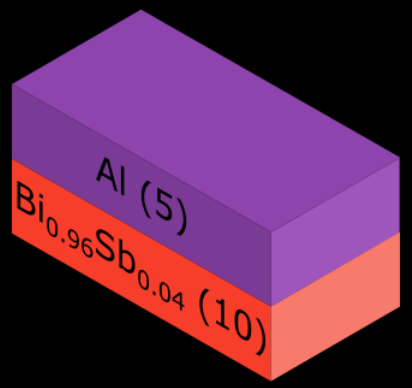
Planar Hall Effect

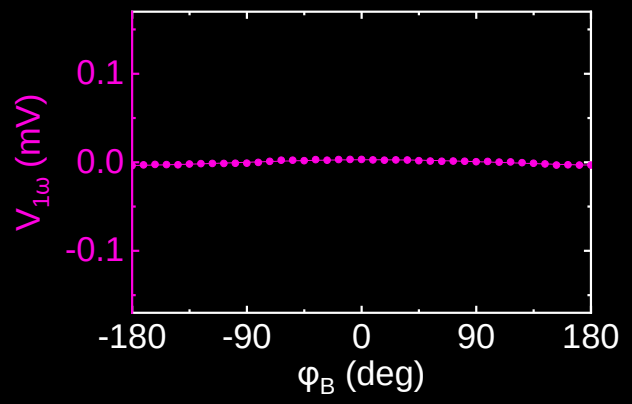
BiSb/Co



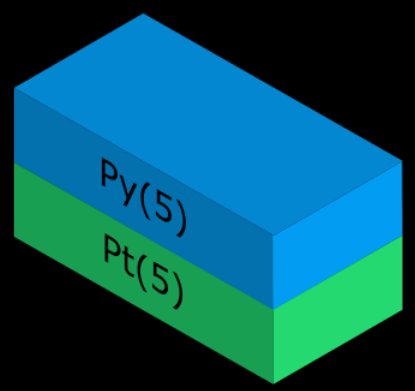


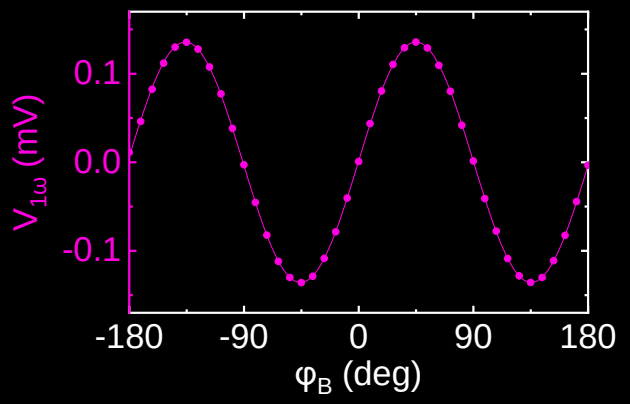
BiSb/Al



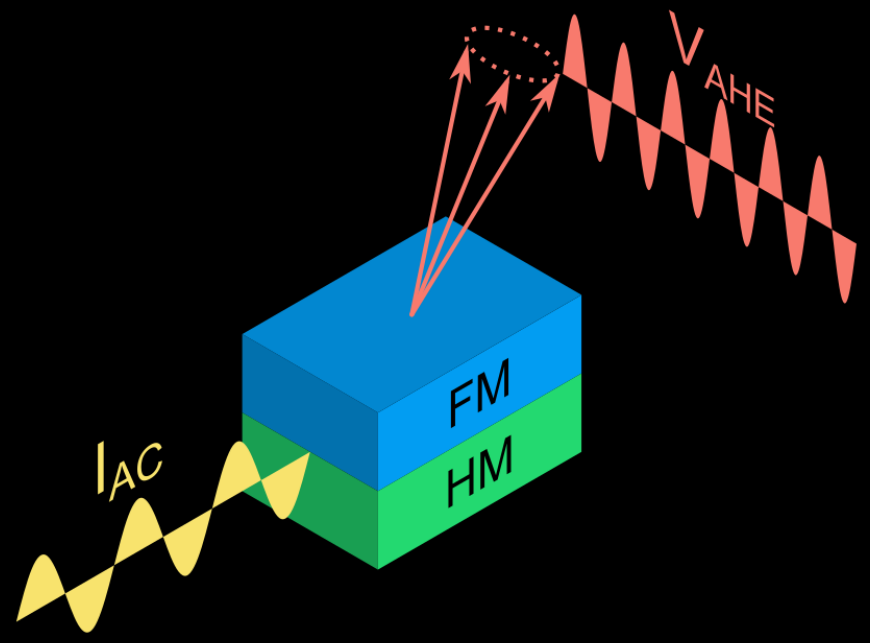


Pt/Py





Second Harmonic Hall voltage



$$V_{xy}^{2\omega} = \left(\frac{V_{\mathrm{AHE}}}{2} \frac{\Delta B_{\mathrm{DL},y}^{\mathrm{IP}}}{B_{\mathrm{ex}} + \mu_0 M_{\mathrm{eff}}} + \alpha \nabla T + \beta \nabla T \cdot B_{\mathrm{ext}} \right) \cos(\varphi_{\mathrm{B}}) + \left(V_{\mathrm{PHE}} \frac{\Delta B_{\mathrm{FL},y}^{\mathrm{IP}}}{B_{\mathrm{ext}}} \right) \cos(2\varphi_{\mathrm{B}}) \cos(\varphi_{\mathrm{B}})$$

Damping-like Torque

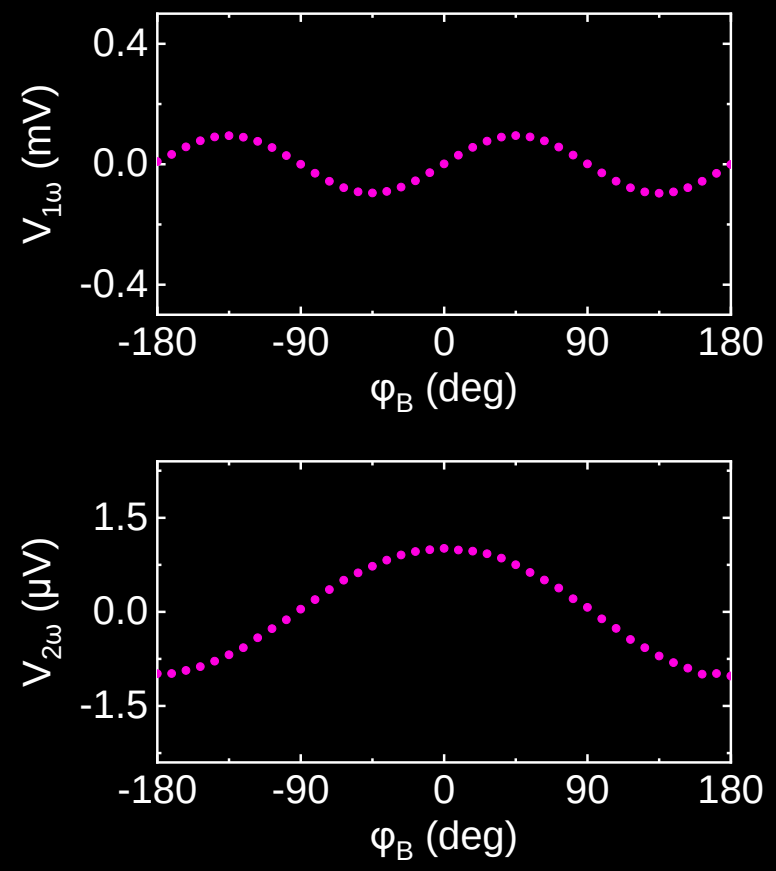
anomalous Nernst Effect

ordinary Nernst Effect

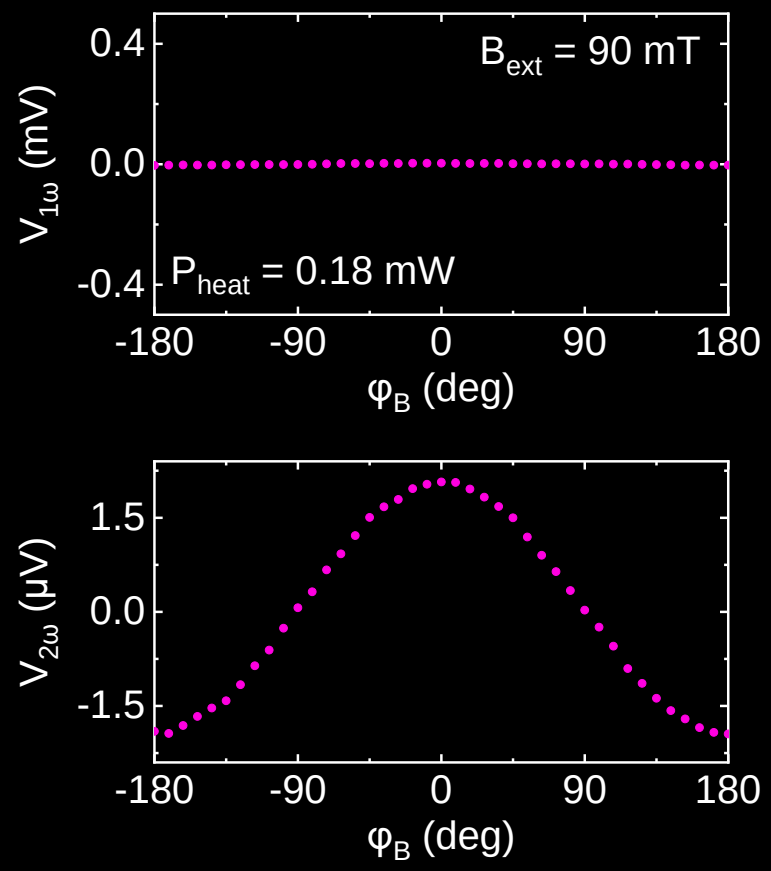
Field-like Torque

Planar Hall Effect

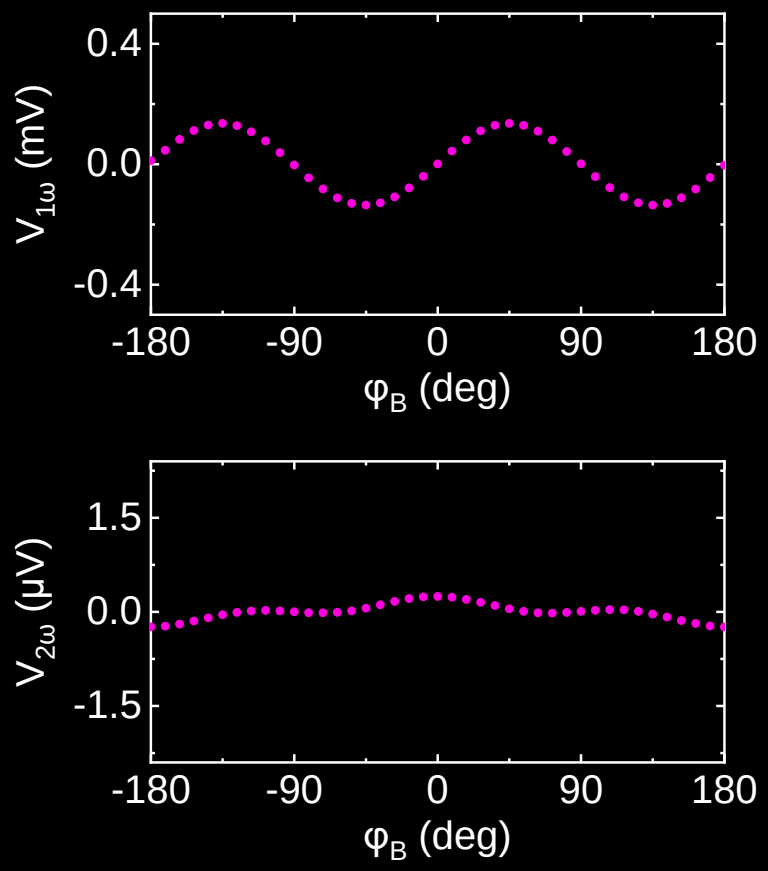
BiSb/Co



BiSb/Al

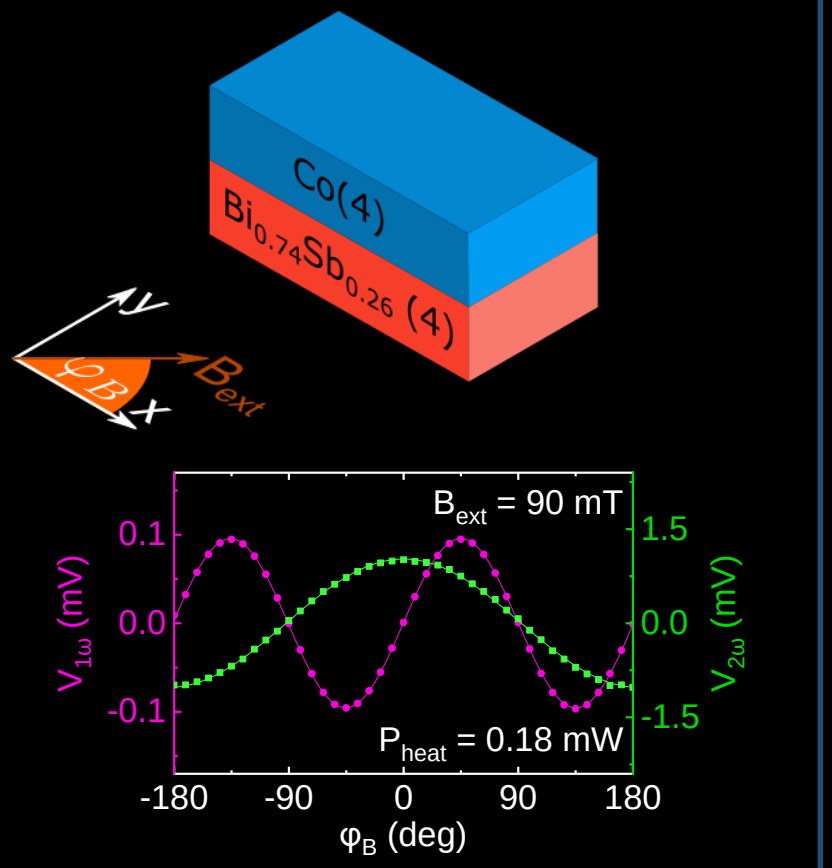


Pt/Py



Planar Hall Effect

BiSb/Co



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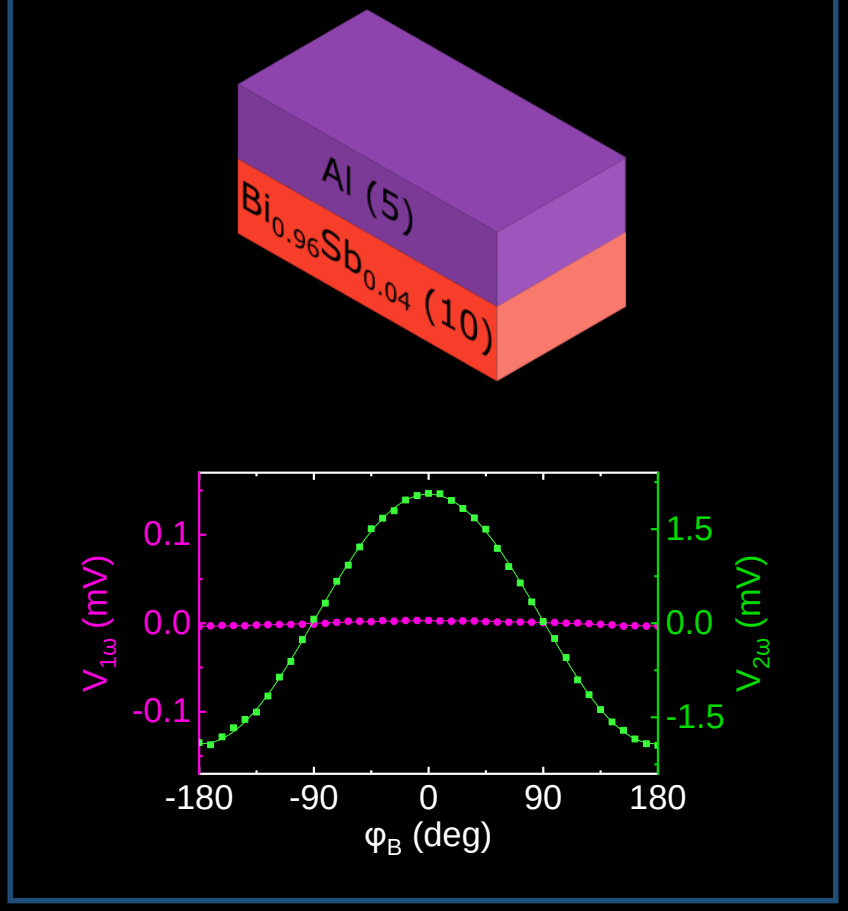
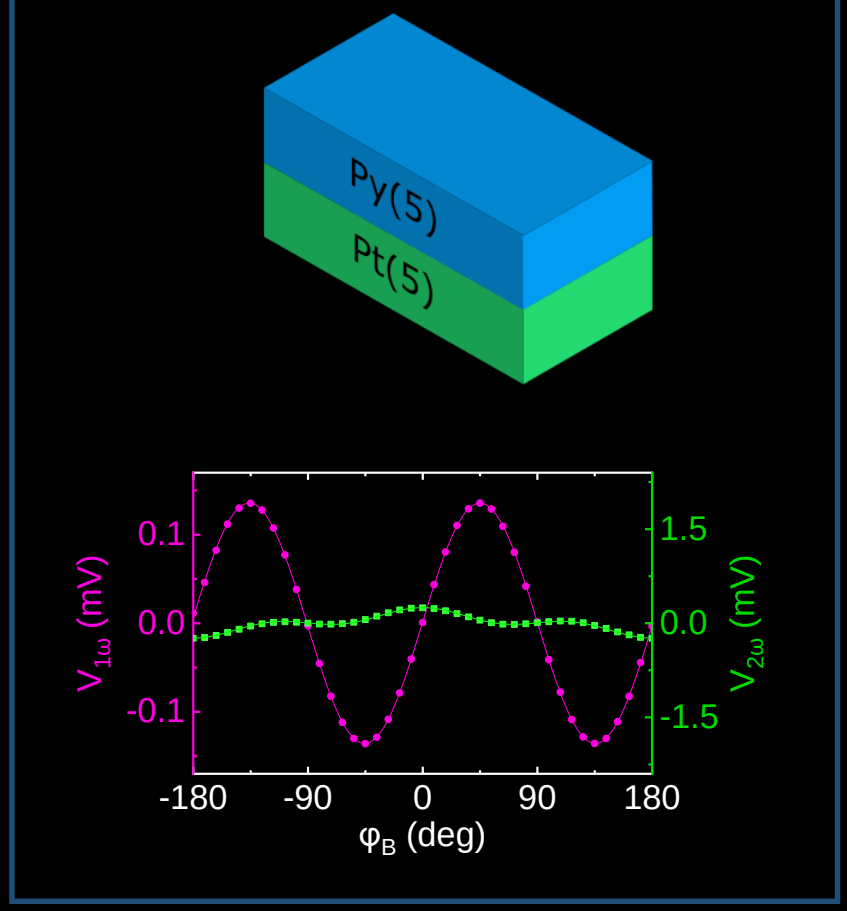
Pt/Py